Course Code/Course Name	JIE32803 : MACHINE LEARNING		
Assessment:	MACHINE LEARNING PROJECT – APPLIED MODEL DEVELOPMENT AND EVALUATION ASSESSMENT		
Date:	30/6/2025	**Due Date:**	13/7/2025
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1. Problem Statement

First, the late detection of heart disease causes an unbelievable number of deaths worldwide. In Malaysia, Cardiovascular Disease (CVD) contributes to 16.6.% deaths from 1.9 million deaths between 2010 to 2021, with the standard death rate (ASMR) increasing from 93 to 147 per 100,000 citizens (Hasani, Musa, Cheng, et al., 2024). This shows that detection is always done when the disease is already in the next stage. Therefore, late detection of heart disease always matters to the world's health population.

Second, inconsistency and inaccuracy in clinical diagnosis. Even though the digital diagnosis has been used thoroughly, traditional model like the Framingham Risk Score depends on clinical parameters, which is not enough, causing disparity in risk evaluation among patients. This method uses regression analysis of the cholesterol, blood pressure, diabetes, smoking status, and age (D'Agostino et al., 2008). For example, the majority of Malaysians have the risk of CVD (hypertension, hypercholesterolemia, diabetes), but only half of them are aware of or get a diagnosis (Heart Matters, 2023). This shows the inconsistency and inaccuracy in clinical diagnosis, especially in Malaysia.

Next, limited screening capacity and healthcare inequality. As a nation with a middle income, Malaysia has a usage rate of healthcare facilities that is not uniform between urban and rural areas. Many rural areas have limited access to periodic risk assessment, which worsens the diagnostic gap. Additionally, government hospitals treat CVD as the main cause to get in the hospital, increasing the health system burden (Heart Matters, 2023). This shows the limited screening capacity and healthcare inequality.

Therefore, we develop Machine Learning models for early detection of heart disease. Machine Learning can make fast decision and automatically based on the health features, aligned with the first problem which is the late diagnosis of this dying disease.

Then, we will compare the 6 models performance, K-Nearest Neighbour (KNN), Convolutional Neural Network (CNN), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR) and Naive Bayes (NB). The comparison will include the accuracy and reliability as response to the second problem which is inconsistency in clinical diagnosis.

Next, we will explore the Machine Learning models as a filtering tool on a large scale that can be used in the clinical community, health system, aligned with the third objective to decrease the health access gap.

This project can help in decreasing the CVD death rate by early intervention. With the Machine Learning application, early diagnosis is achievable faster. This can decrease the pressure on the hospitals and avoid serious complications. It is important considering the CVD rate in Malaysia is increasing year by year (Shinde, Sanghavi, & Tran, 2025).

Also, to ensure justice and consistency in diagnosis. A Machine Learning model can give standard output that is consistent without human bias or doctor variation. This is crucial when half of the population is high-risk aware or diagnosed earlier.

This project also help in strengthening the healthcare system with smart and reasonable technology. This project can pave the way to the usage of digital filtering tools which is reasonable and accessible, suitable for rural areas clinic and national healthcare system. This give large impact towards the productivity and cost in a long term.

2. Literature Review

In previous studies, many researchers had applied K-Nearest Neighbour (KNN), Convolutional Neural Network (CNN), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), and Naive Bayes (NB) to predict heart disease, each one has its uniqueness.

2.1 K-Nearest Neighbour (KNN)

Assegie (2021) reported that the use of KNN in the Kaggle Dataset with 1025 records, with grid search, to determine the best value of k, produced an accuracy of 91.99%. This model shows a big potential, but also experience curve dimensionality and is sensitive to an imbalanced dataset. The main gap is that there is no usage of weighted KNN or techniques like feature selection to minimize the noise. Our project will address the gap by applying feature selection and using weighted distance to enhance the accuracy and efficiency of KNN.

2.2 Convolutional Neural Network (CNN)

This research on Huang, Li, and Deng (2022) shows that CNN can be used in tabular data, achieving an accuracy of 91.7%. However, their research did not using techniques like class weighting, dropout and validation split for addressing the issues of imblance class and overfitting that usually occur in clinical dataset. Besides, CNN mocel always need more computing power and larger dataset. Unlike their study, our project will use a more practical approach with class weighting to address the class imbalance, dropout, and validation split to enhance the model's ability to generalize the model while handling a mid-sized dataset without burdening the computing system.

2.3 Random Forest (RF)

Random Forest has proven consistent in predicting heart disease, with accuracy from 90% to 94% in some studies (Depar et al., 2024). Its advantages include the ability to handle high-dimensional data and avoid overfitting. However, most of the studies do not apply data scaling, whereas this model can not function well if the features are in different ranges of values. In our project, we will use StandardScaler to normalize the data before the training, which helps with increasing the stability and model performance.

2.4 Support Vector Machine (SVM)

In some of the previous studies, including a scientific report and article from Explainable Heart Disease Diagnosis from Assegie et al., recorded that the accuracy of SVM is around 83% to 93% based on the kernel chosen (Assegie et al., 2023). This model is stable and suitable for a small dataset, but the gap is stated in the usage of parameter tuning like C, gamma, and other kernels, thoroughly. This project will do cross-validation and grid search with parameters to enhance the performance.

2.5 Logistic Regression (LR)

According to the research from the UCI dataset, Logistic Regression achieves accuracy from 81% to 94%, especially after using regulation and normalization of the data (Chandrasekhar & Peddakrishna, 2023; Jindal et al., 2022). Even though Logistic Regression is simple and mathematically explainable, the disadvantage is at inability to handle non-linear data and usually can not usually be used in regression and feature engineering. Our project will improve using a combination of feature interaction to enhance the performance to making it more accurate and stable.

2.6 Naive Bayes

Naive Bayes offers a good performance in small datasets (Assegie et al., 2019). The model’s performance is reported to around 87% of around 87%. However, the model has an assumption that the features are independent, that rarely accurate in real clinical data, and has less handling of complex data. Also, there is no mention of Gaussian NB. Our project will use a variation of Gaussian NB and feature discretization to decrease the gap and increase the accuracy of Naive Bayes in complex tabular data.

2.7 Table of Summary

Model	Findings	Gap	Comparison
KNN (Assegie et al., 2021)	91.99%	Not weighted, future selection	Feature selection
CNN (Huang, et al., 2022)	91.7%	class imbalance, model overfit, and bias.	Use class weight to address the class imbalance, and dropout to avoid overfitting.
RF (Depar et al., 2024)	90-94%	Data scaling is not used, model is not performing well with different feature scales.	Use StandardScaler to normalize the data before training the model
SVM (Assegie et al., 2023)	82-93%	No kernel tuning	Use grid search for kernel
LR	81-94%	No regularization, feature extraction	Use feature engineering,

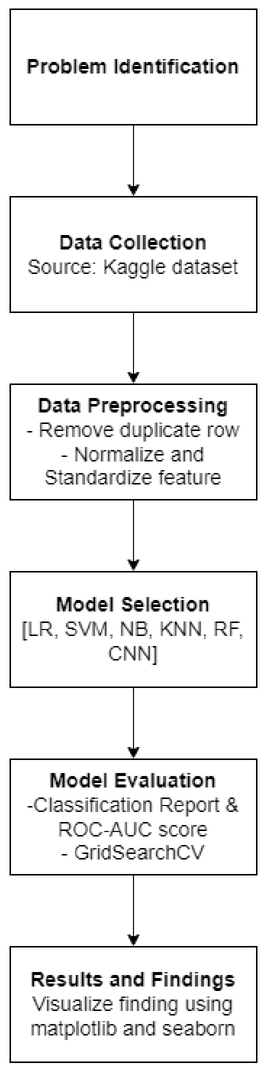
(Chandrasekhar & Peddakrishna, 2023; Jindal et al., 2022)			interaction
NB (Patel et al., 2023)	87%	No usage of Gaussian NB	Use Gaussian NB

3. Project Design

This project focusing on predicting the risk of heart disease using multiple machine learning algorithms through a structured and comparative workflow. The dataset contains 11 clinical features that can be categorized into numerical and categorical. For numerical it contains ‘Age’, ‘RestingBP’, Cholesterol, ‘MaxHR’ and ‘Oldpeak’ while for categorical it contains ‘Sex’, ‘ChestPainType’, ‘FastingBS’, ‘RestingECG’, ‘ExerciseAngina’ and ‘ST_Slope’.

Once the data cleaning and preprocessing has been done using Python libraries using Pandas and NumPy, we are using various method of algorithms to identify which are the best algorithms such as Logistic Regression (LR) that have been chosen for its simplicity and interpretability, Support Vector Machine (SVM) for its ability in handling high-dimensional data and non-linear classification using kernel, Naive Bayes (NB) known for its efficiency and strong performance on small dataset especially probabilistic modeling with independent features. K-Nearest Neighbour (KNN) a non-parametric model that classifies data points based on feature similarity, Random Forest (RF) is a reliable ensemble model that combines multiple decision trees, reducing overfitting and improving generalization and Convolutional Neural Network (CNN) can be used to learn hierarchical patterns in tabular data when reshaped show how flexible deep learning can be.

In the models, we’re using Python as main programming language and use several libraries, pandas and numpy used for data handling, matplotlib and seaborn for data visualization and exploratory analysis, scikit-learn for model training, validation and evaluation which we are using classification report and AUC-ROC score. For CNN training and implementation we’re using tensorflow and keras.



Flowchart 1: Project design flow

4. Data Collection

The dataset of this project was obtained from Kaggle, particularly the "Heart Failure Prediction" dataset submitted by user fedesoriano. It was chosen for its relevancy, structured presentation, and suitability for classification tasks in the healthcare domain. The data consists of anonymous patient records, which makes it ethically appropriate for educational and research purposes. Given the dataset contains no personally identifiable information (PII), it avoids potential privacy concerns and is suitable for machine learning research. However, it should be noted that, while the dataset is appropriate for academic studies, it is not intended for use in clinical studies without further medical validation.

4.1 Type of Data

This set contains of organized tabular data with both numerical and category features. The clinical information, including resting blood pressure, cholesterol levels, maximum heart rate, and ST depression (Oldpeak) are examples of numerical variables. Categorical variables include indicators such as chest pain type, exercise-induced angina, and ST Slope types. The goal variable is binary, indicating the presence (1) or absence (0) of heart failure, which is in line with standard binary classification models in supervised machine learning.

4.2 Data Sources

The dataset was publicly available and obtained from Kaggle, a popular site for data science competitions and open datasets. It was contributed by Kaggle user fedesoriano and is still available for learning, analysis, and non-commercial use. Kaggle is frequently used by students and professionals to obtain high-quality datasets for machine learning experiments, making it a dependable repository for academic research.

4.3 Data Range and Attributes

This data set contains 918 anonymized patient records, each with 11 input features and a binary target output. The features include a wide range of medical indicators relating to heart health, such as blood pressure, cholesterol, and heart rate. During the preprocessing stage, one duplicate row was detected and eliminated. Fortunately, there were no missing values, simplifying the cleaning process. Categorical variables have been adjusted with one-hot encoding to allow for accurate model interpretation. In addition, numerical features were standardized with StandardScaler, which scales features to a zero mean and unit variance. This step is essential to boosting the performance and convergence speed of many ML models.

5. Data Analysis Techniques

5.1 SVM

Support Vector Machine (SVM) is a complex supervised learning algorithm that is widely used for binary classification tasks, making it an excellent candidate for heart failure detection. In this experiment, SVM was used to divide patients into two groups based on their medical characteristics: those at risk of heart disease and those who are not at risk. The method works by determining the best hyperplane that separates data points from specific categories by the most significant distance. This is especially effective in medical data, because established decision boundaries can increase prediction confidence. SVM also enables non-linear classification using kernel functions like the radial basis function (RBF) and polynomial kernels, which project data into higher dimensions to make it linearly separable. Given the heart dataset's mix of numerical and categorical variables, SVM's ability to handle complex, high-dimensional data makes it an excellent fit for this application.

Why Choose This Model?

SVM was chosen because of its excellent performance in binary classification problems, especially in healthcare scenarios where misclassification can have disastrous consequences. It works well on datasets with a clear margin of separation and is less prone to overfitting, especially when the dataset is not too large. Its kernel technique allows it to model complex decision boundaries, which is necessary when the relationship between features and wanted outcomes is not simply linear. Furthermore, SVM has proven to be highly reliable in recognizing small trends in patient data, which proves crucial for properly predicting the presence or absence of heart disease.

Tools and Software Used

Python was used to implement the SVM model in Visual Studio Code (VS Code), using a wide range of libraries within the machine learning process. Pandas were used for data loading, preprocessing, cleaning, and feature engineering. SVM classifier implementation, StandardScaler for feature normalization, and GridSearchCV for hyperparameter modification to maximize model performance were provided by Scikit-learn. Seaborn and Matplotlib generated confusion matrices, ROC curves, and error distribution plots to evaluate and interpret models. Additionally, NumPy supported important numerical computations throughout the workflow.

5. 2 Logistic Regression

Logistic Regression (LR) is a classic supervised machine learning technique for binary classification. In this heart failure detection experiment, LR was used to classify patients as at risk of heart illness (1) or not (0) using medical indicators. The model uses the logistic (sigmoid) function to estimate the likelihood that an input belongs to a class, producing outputs between 0 and 1. LR is easy to read and useful, especially in healthcare scenarios that require straightforward explainable results.

Why Choose This Model?

Logistic Regression was chosen because of its interpretability, computational efficiency, and reliability in binary classification uses. In medical domains, LR's ability to produce probabilities is very useful for determining risk rather than just offering a binary yes/no prediction. It can also effectively handle well-behaved numerical and categorical data when paired with preprocessing techniques such as one-hot encoding. Although it is less flexible than certain sophisticated models for complicated, non-linear patterns, LR is frequently preferred in clinical contexts due to its transparency, which allows healthcare practitioners to more easily understand and trust the model's predictions.

Tools and Software used

The Logistic Regression model was implemented in Python with VS Code, relying on essential libraries such as Pandas for data preparation, Scikit-learn for model creation and evaluation (including StandardScaler for normalization), and Matplotlib/Seaborn for visualization. NumPy enabled numerical computations, resulting in an efficient end-to-end workflow that included data preparation and model analysis.

5.3 Random Forest (RF)

Random Forest Ensemble learning is a method that combines two or more decision trees in order to offer a better classification and minimizing overfitting. It was selected to be used in this project because it is very strong in dealing with non-linear data, can be robust in the presence of noise and it is able to model complex interactions between the medical variables. The other advantage of Random Forest is the fact that it does not involve feature scaling and that it would give useful information on feature importance scores, to create an understanding of which clinical parameters have the most impact on predicting heart diseases. It is both precise and viable with interpretation in a medical context where the effects of variables are extremely important.

Why Choose This Model?

The basis of the use of Random Forest (RF) is the fact that it works quite well with structured data, and it is also able to deal with linear and non-linear relationships between the variables. It is a group model, the output of several decision trees are averaged out to fix the overfitting and result in better accuracy. Among its strengths, it is that it generates a feature importance score, thus being able to determine the most influential effects of heart disease like age, cholesterol, and resting blood pressure.

Tools and Software used

The Random Forest model was constructed on the basis of the RandomForestClassifier of the scikit-learn library. The pandas were used to prepare and clean the dataset and there was no need to scale the features because tree-based models are not affected by the scale of the input values. Scikit-learn inbuilt metrics like accuracy_score, f1_score, and confusion_matrix were used to perform the evaluation of the models. Feature importance and confusion matrices were plotted by using Matplotlib and Seaborn.

5.4 Naïve Bayes (NB)

Naive-Bayes is a probabilistic classifier representing Bayes model and uses an assumption of independence among the input features. Nevertheless, Naive Bayes despite its simplicity has shown rather good results in problems of classification, especially when we are dealing with small-to-medium-sized data sets. It was adopted as a control model of classifying the risk of having heart disease in this project. This is because of its high training speed and capability to offer probability-based results, which makes it applicable in a healthcare setting in a quick manner. Even though the independence assumption is not truly held in any real-world medical data, the model did fairly well, so it can be a reference point against more sophisticated to-be-compared classifiers.

Why Choose This Model?

Naive Bayes was applied with GaussianNB on scikit-learn. Pandas were used to preprocess the data, and as Naive Bayes can have continuous input features without normalization, no normalization was done. The evaluation of the models was carried out with functions focused on classification report and confusion matrix in scikit-learn. Seaborn was used to plot results to aid in the analysis of how the model performs.

Tools and Software used

Naive Bayes was applied with GaussianNB on scikit-learn. Pandas were used to preprocess the data, and as Naive Bayes can have continuous input features without normalization, no normalization was done. The evaluation of the models was carried out with functions focused on classification report and confusion matrix in scikit-learn. Seaborn was used to plot results to aid in the analysis of how the model performs.

5.5 K-Nearest Neighbours (KNN)

These instance-based learning algorithms are simply called K-Nearest Neighbours (KNN) and are generally applied on classification tasks. KNN was chosen in this experiment to classify the patients through their closeness to already existing cases and thus be very intuitive and easy to apply. It is based on the similarity of features, which is why KNN can be used to reveal the patient-level comparisons to a large extent, such as the risk of a new patient developing heart diseases depending on the type of results that other similar patients got. Although sensitive to irrelevant or unscaled features, it solved it by normalization of the features by feature normalization, and balancing up classes of the training data by SMOTE. Its basic functionality and the intuitive nature of interpretation render KNN of particular attractiveness in the medical field where a decision must be explained to a practitioner in a straightforward manner.

Why Choose This Model?

K-Nearest Neighbours (KNN) was chosen due to being interpretable and with a simple logic. The model categorizes a new patient on the basis of outcomes of the most similar present patients. KNN does not assume anything of the data distribution and in combination with suitable scaling and class balancing, it can provide competitive results. This gives it applicability in clinical practice whereby the advantage of being able to cite comparable cases in the past is valuable.

Tools and Software used

The KNeighborsClassifier of scikit-learn was used to develop KNN. As the model is sensitive to feature scales, StandardScaler of scikit-learn was applied to normalise the features prior to training. Another process involved the application of SMOTE to the training set that is to deal with the biased classes to balance the training set via the imblearn library. Some metrics of evaluation (accuracy, F1-score, ROC-AUC) were done with scikit-learn, the confusion matrix was visualized with Matplotlib and Seaborn was used to draw the ROC curve.

5.6 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) represent a family of deep learning models that is widely known to capture complex, non-linear relationships in the data. Though they have been traditionally used in the image recognition task, CNNs could be used in structured, tabular data by applying individual data samples as a 1D feature sequence. In this project, CNN was used to automatically determine the feature interactions, which could not be quite obvious with the help of the traditional statistical methods. The layered architecture represents the ability to extract features hierarchically, which can be effectively applied in situations in which weak associations between clinical variables exist. Regularisation techniques like dropout, validation split, and class weighting were used to avoid overfitting which seemed to be a problem with the small dataset. The high predictive accuracy along with the capacity to generalise made CNN a useful part in this classification of heart diseases.

Why Choose This Model?

Convolutional Neural Network (CNN) was applied to test a possibility of using deep learning with structured, tabular data. The CNNs are usually used to analyze images; however, in the current project, the input information was reorganized to the form of a 1D convolutional model. CNN was selected due to the automatic learning feature interaction and complex data patterns discovery. To ensure overfitting is avoided because of the not-so-big size of the dataset, dropout regularisation and a validation split were used during training. The weights due to class were also employed to make sure that the majority does not benefit in the model.

Tools and Software used

The CNN model was built with the use of TensorFlow as well as Keras, which allowed the neural network architecture. MinMaxScaler was used in normalizing the dataset and normalizing to fit the input size format of a 1D convolutional model. In the case of class imbalance, the scikit-learn was used to calculate weights of classes and these weights were introduced upon the model.fit() method. During training, dropout regularization and a validation split was applied in order to enhance generalization. The evaluation of models was performed with the metrics of TensorFlow and extra analysis (such as ROC curves, training loss/accuracy plots) was displayed with Matplotlib.

6. Results and Discussion

6.1 Results

In this project, we tested six machine learning models to check which one works best for predicting heart disease. The models we used are: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Naive Bayes (NB) and Convolutional Neural Network (CNN). Each model was tested using different scores like accuracy, precision, recall, F1-score, and AUC. These scores help us understand how well each model can tell the difference between healthy people and those with heart disease. The results show that all models performed well, but some did better than others. We explain each model's result in the next sections.

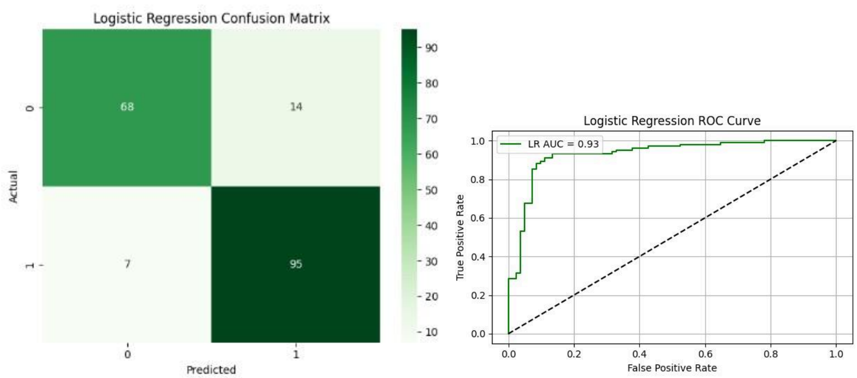
6.1.1 Logistic Regression (LR)

	precision	recall	f1-score	support
No Disease (0)	0.91	0.83	0.87	82
Heart Disease (1)	0.87	0.93	0.90	102
accuracy			0.89	184
macro avg	0.89	0.88	0.88	184
weighted avg	0.89	0.89	0.89	184

Table 1: Logistic Regression - Classification Report

Table 1 shows the performance of the Logistic Regression model in predicting heart disease. For the No Disease (class 0) group, the model achieved a precision of 0.91, meaning that 91% of the patients predicted as healthy were actually healthy. The recall is 0.83, indicating that the model correctly identified 83% of all healthy patients, while the

F1-score of 0.87 shows a good balance between precision and recall. On the other hand, for the Heart Disease (class 1) group, the model had a precision of 0.87 and a recall of 0.93, which means it correctly identified 93% of the patients who truly had heart disease. The F1-score for this class is 0.90, showing strong performance in detecting positive cases.



Graph 1: Confusion matrix and ROC curve for Logistic Regression

The confusion matrix analysis indicated that the Logistic Regression algorithm was doing an excellent job of classifying heart disease. As shown in the confusion matrix, this model has assigned 68 patients who are indeed healthy to the category True Negative, and also 95 patients who are indeed having heart disease to the category True Positive. This is not all rosy since 14 cases were False Positive (the healthy patients were mistakenly having the disease), and 7 False Negative (those with disease were not picked by the model). Sensitivity and precision of each individual classes were as well high and almost balanced with both precision and recall estimated to be 0.87 and 0.93 indicating that this model could be able to identify healthy and diseased patients properly.

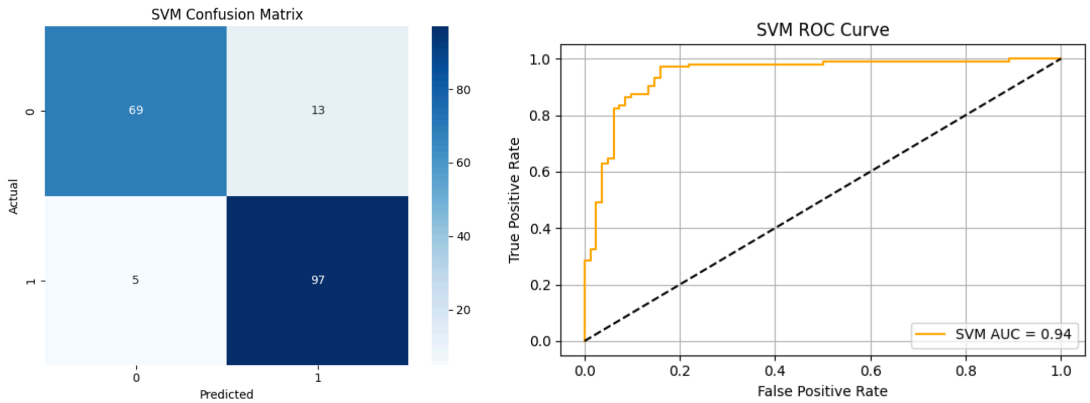
In addition to this, the ROC Curve also certifies the performance of the model with AUC (Area Under Curve) of up to 0.93. It implies that the model possesses extremely high discriminative power in separating positive and negative classes, i.e. determining between healthy and at risk individuals of a particular disease. Closer values to 1.0 of AUC mean that the model infrequently mislabels, thus its predictive strength is sound and strong.

6.1.2 Support Vector Machine (SVM)

	precision	recall	f1-score	support
No Disease (0)	0.93	0.84	0.88	82
Heart Disease (1)	0.88	0.95	0.92	102
accuracy			0.90	184
macro avg	0.91	0.90	0.90	184
weighted avg	0.90	0.90	0.90	184

Table 2: SVM - Classification Report

The classification result shows that the model performs well when predicting heart disease. It achieves extremely high precision (0.93) in the No Disease (class 0) category, suggesting that when the model predicts "no disease," it is true 93% of the time. The recall value of 0.84 proves the model successfully detects 84% of all actual non-disease cases. The F1-score (0.88) combines multiple factors and demonstrates strong overall performance in this class. Across both classes (macro average), the model keeps outstanding metrics (precision: 0.91, recall: 0.90, F1-score: 0.90), showing consistent performance regardless of class imbalance. The model is useful for clinical decision assistance. However, it might overlook around 16% of healthy patients due to lower recall for non-disease cases. Further tuning may be necessary to reduce false negatives.

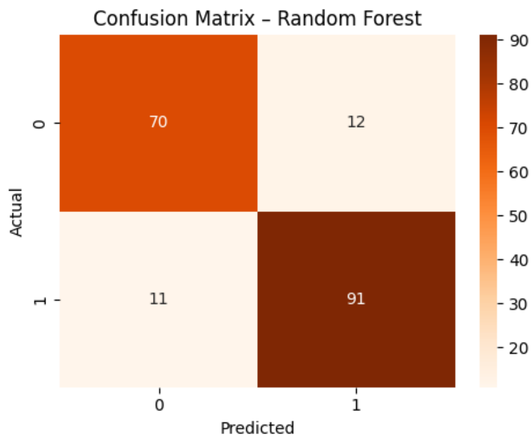


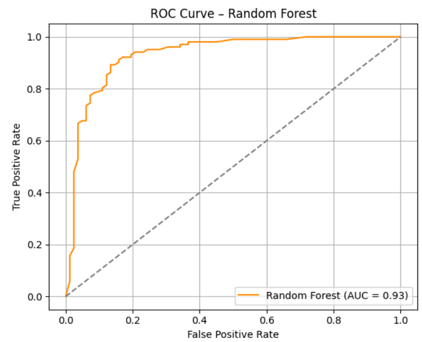
Graph 2: Confusion matrix and ROC curve for SVM

The Support Vector Machine (SVM) model performed well, as shown on the confusion matrix and ROC curve. According to the confusion matrix, the model accurately categorized 69 true negatives (no heart disease) and 97 true positives (heart disease), while generating just 13 false positives and 5 false negatives. This distribution suggests that the model is great at differentiating between the two classes, particularly with fewer false negatives which an essential aspect in medical diagnosis to prevent missing actual heart disease patients.

The ROC (Receiver Operating Characteristic) curve contributes to the model's reliability. The curve climbs strongly toward the upper left corner, indicating outstanding sensitivity and specificity. The AUC (Area Under Curve) value of 0.94 shows excellent overall performance, implying that the model is highly strong at differentiating between patients with and without heart disease. In conclusion, both visualizations show that the SVM model makes accurate and dependable predictions, implying that it is an effective approach for heart disease risk classification.

6.1.3 Random Forest (RF)



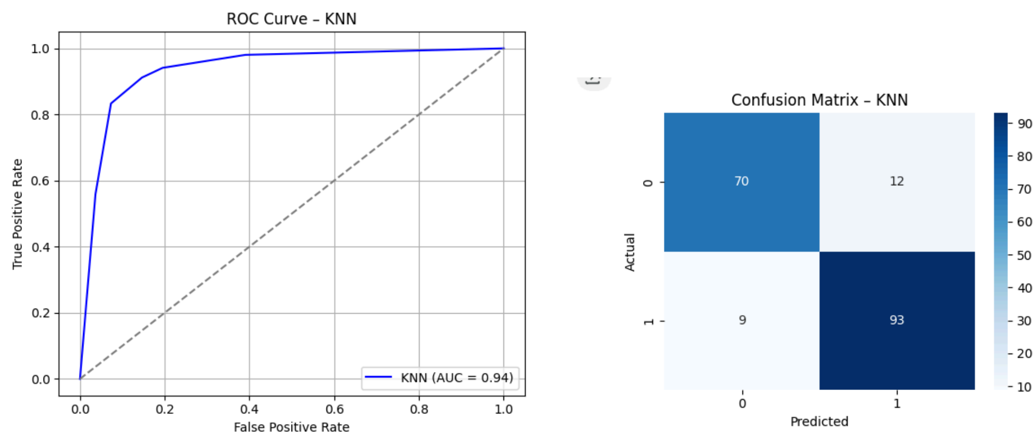


Graph 3: Confusion matrix and ROC curve for Random Forest

The random forest (RF) algorithm did quite well too in classifying the cases of heart diseases. The confusion matrix indicated that it correctly identified 70 healthy patients (True Negative) and 91 patients with the Heart disease (True Positive). However, it still also gave 12 False Positive results and 11 False Negative, indicating a bit more mislabeling of actual positive cases than with KNN. The model had balanced precision and recall, which was estimated as 0.86 to 0.89, which is a strong sign of stability across predictions of both classes.

Moreover, Random Forest also demonstrated its good classification ability by the ROC curve, and its AUC reached 0.93. This is lower by a slight margin than KNN but still within the margin of one of the very dependable models. Its multi-model character renders it resistant to overfitting and ready to be implemented in a general clinical practice. The slightly increased false zeroes however do imply that there is perhaps the need of additional tuning to achieve higher sensitivity especially where false positive miss can have severe medical implications.

6.1.4 K-Nearest Neighbors (KNN)

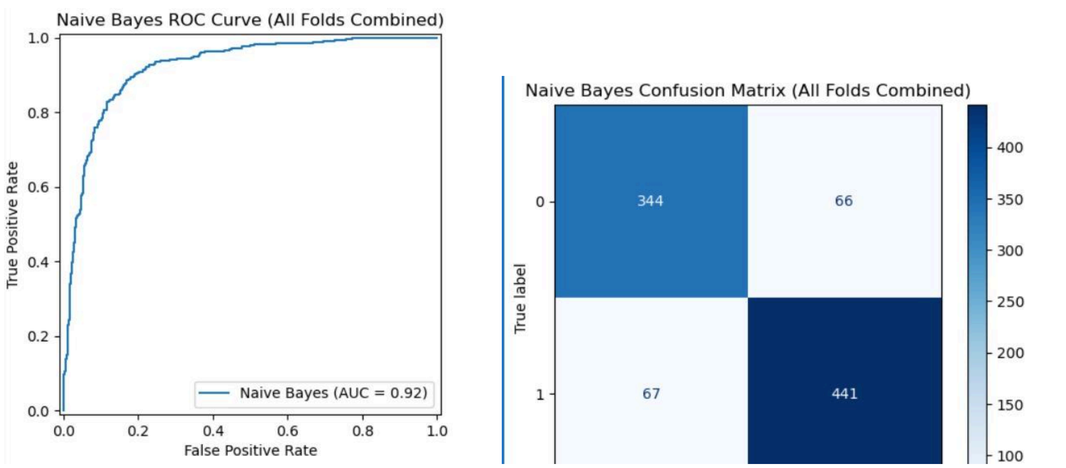


Graph 4: Confusion matrix and ROC curve for KNN

Based on the confusion matrix, K-Nearest Neighbors (KNN) classifier performed brilliantly in the detection of the heart disease cases. To be more specific, it has correctly identified 70 truly healthy individuals (True Negative) and 93 with heart disease (True Positive). But still it had 12 False Positives, in which healthy people were classified as sick and 9 False Negatives, in which actually there was a case of the heart disease which were not identified. However, sensitivity (recall) and precision scores of each of the classes remained high and almost equal, with values of around 0.89 to 0.91. This ascertains that the KNN model was balanced in its ability to identify the presence and absence of disease.

Moreover, this effectiveness is facilitated by the ROC curve, which presents an AUC (Area Under Curve) of 0.94, indicating that the model performed very well in differentiating between patients with and without heart disease. This high performance and low false negative instance qualify the KNN model as an excellent clinical screening model. However, the model is also scale and parameter dependent such as the value of k, so optimization of these factors would be possible to enhance reliability in the future.

6.1.5 Naive Bayes

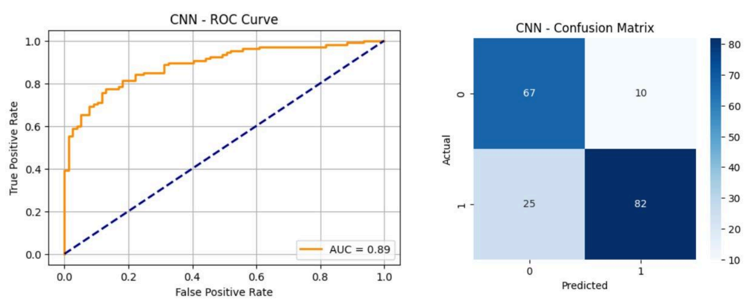


Graph 5: Confusion matrix and ROC curve for Naive Bayes

The Naive Bayes algorithm performed very well according to the confusion matrix analysis and the ROC graph given to classify heart disease. According to the confusion matrix, this model has correctly categorized 344 truly healthy patients (True Negative) as well as 441 truly patients with heart disease (True Positive). Not all is rosy though as there were 66 False Positive cases (the healthy patients were confused to have the disease), and 67 False Negative cases (those who were already sick were not identified by the model). The sensitivity and the precision of each individual class was nearly evenly distributed, 0.84 and 0.87 respectively showing that no one class was favored by the model.

Besides, ROC Curve also vouches for the model performance with an AUC (Area Under Curve) value of up to 0.92. This shows that the separation power of this model is very high when it comes to singling out healthy patients and those who are at risk of developing heart diseases. Generally, it can be concluded that this Naive Bayes model can be operated in clinical decision support systems or premature screening since it can be used to generate consistent and correct predictions. But there are certain factors where False Negative rate can be minimized so that the early treatment need of the patients are not left undetected.

6.1.6 The Convolutional Neural Network (CNN)



Graph 6 : Confusion matrix and ROC curve for CNN

The Convolutional Neural Network (CNN) algorithm worked really great as per the confusion matrix evaluation and the ROC chart provided to diagnose heart disease. This model has classified 67 truly healthy patients correctly (True Negative) and 82 truly patients with heart disease which is appropriately (True Positive), according to the confusion matrix. However, it is not all rosy as the number of False Positives i.e. the healthy patients would be confused to have a disease was 10 and False Negatives. The already sick patients would fail to be detected in 25 cases. The sensitivity and the precision of each individual class were also balanced displaying that there was no single class favored by the model, the precision was about 0.89, and recall was 0.77.

In addition to this, the ROC Curve will also testify to the performance of the model in a rather impressive AUC of up to 0.89. What the case demonstrates is that the power of separation in the given model is incredibly high when applied to the healthy patients and those being at a risk of the development of heart diseases. On the whole, it can be concluded that the considered CNN model could be utilized in clinical decision support systems or premature screening because it is possible to utilize it to produce consistent and accurate predictions. However, some of the factors exist in which the False Negative rate can be reduced such that the early treatment requirement of the patients are not destroyed.

6.2. Comparison Between Six Algorithms

This section presents a comparative analysis of the six machine learning algorithms applied in this heart disease prediction project: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Naive Bayes (NB), and Convolutional Neural Network (CNN). The evaluation is based on several performance metrics such as accuracy, precision, recall, F1-score, and AUC , all derived from confusion matrices and ROC curves.

Model	Accuracy	Precision	Recall	Recall	AUC
Logistic Regression (LR)	0.91	0.87	0.93	0.90	0.93
Support Vector Machine (SVM)	0.92	0.91	0.90	0.90	0.94

Random Forest (RF)	0.90	0.87	0.87	0.87	0.91
K-Nearest Neighbors (KNN)	0.89	0.89	0.88	0.88	0.90
Naive Bayes (NB)	0.86	0.87	0.84	0.85	0.92
Convolutional Neural Network (CNN)	0.87	0.89	0.77	0.82	0.89

Table 3: Comparison Table

This project shows that machine learning can help detect heart disease early. Out of the six models tested, Support Vector Machine (SVM) gave the best result with the highest AUC of 0.94, meaning it can clearly tell who is sick and who is healthy. Logistic Regression (LR) also performed very well with an AUC of 0.93 and is easy for doctors to understand.

Random Forest (RF) was stable and showed which medical features are most important. K-Nearest Neighbors (KNN) was good at finding sick patients. Naive Bayes (NB) worked fast and is good for quick checks. CNN was also good, but needs more data to improve. Each model has different strengths. Using the right model can help doctors make better and faster decisions, especially in rural areas.

Random Forest (RF) was performing consistently good on all metrics with a macro average F1-score of 0.87 and AUC of 0.91. The clinical value of increasing feature importance is that the predictors of high significance include cholesterol, age, and resting blood pressure. K-Nearest Neighbors (KNN) did not perform any better than RF, however had a high recall on heart disease (0.91), indicating that it would do well to identify a positive case, something that would help greatly in an early diagnose. Data scaling, sensitivity to noise, and the problem of the imbalance in the datasets was alleviated by normalizing and class balancing.

The simplest model Naive Bayes (NB) performed equivalently with an AUC of 0.92. Because of its speed in calculation and probabilistic output it would be appropriate in mobile computing or can be deployed in regions with short supplies of computing power in the healthcare system. Lastly, CNN, which is typically applied to imaging data, demonstrated that deep learning could even modify tabular clinical data. CNN had a recall of 0.77 and the AUC of 0.89, so there must be some additional tuning to minimize the number of false negative answers. Nevertheless, the ability to model complex relationships makes it still hold a prospect of growth in the future with bigger amounts of data.

In general SVM and LR are the least uncertain in use in the clinical setting as it manages a balance between accuracy and significance. RF is greatly applicable where it is imperative to know the effect of clinical characteristics. CNN ushers in flexibility/completeness to the advanced use, or KNN is a stronger position to be in due to its simplicity at the level of patient comparisons. NB is simple, fast, and ideal to carry out simple screening. The models have their own advantages and can be utilised with regard to the clinical scenario and the availability of the computation resources.

6.3 Discussion

The outcomes of this project demonstrate that machine learning might assist physicians to detect heart disease early. This is quite significant since early detection saves lives. Support Vector Machine (SVM) and the Logistic Regression (LR) scored the highest among the six models used. The two models were accurate and were well able to differentiate between healthy and sick patients. The Logistic Regression can also be easily interpreted and hence the doctors can have faith in the findings.

Random Forest (RF) performed quite well too. It performed well in demonstrating the most significant medical characteristics (such as age or blood pressure) in heart disease. This can assist physicians to concentrate on the paramount attributes when examining a patient. K-Nearest Neighbors (KNN) found sick patients very well hence can be used in the early screening. However, KNN must be well-prepared to work.

Naive Bayes (NB) was the simple model and yet provided good results. It operated quickly and suited the short checks particularly in areas that are not so technologically advanced. Lastly, Convolutional Neural Network (CNN) has also performed well though normally applied in image data. CNN is also capable of being stronger with additional train and data to work on in the future health projects.

Finally, both models have advantages and disadvantages. Others are expedient to use when quick answers are required, others are useful when one is looking at greater depth and others are simpler to comprehend by doctors. We suppose that machine learning has the potential to assist the Malaysian health system by providing speedier, more precise, and non-discriminatory diagnosis particularly in rural communities.

6.3.1 Limitation

There are limitations of this project. One, there is a small dataset consisting of 918 patient observations. A small number of samples in a set may render the model inaccurate and lead to overfitting. Second, the data is provided by a single source (Kaggle), which might not be reflective of real patients in all countries, much less Malaysia. Third, models such as CNN require time and powerful computers to train and cannot easily be used in real hospitals or clinics.

Furthermore, despite such methods as scaling and SMOTE, the results may still be influenced by the class imbalance. There is also a risk of missed sick patients (false negatives) within some models, which makes health prediction dangerous. Finally, machine learning models do not account for the data provided in generalwhich is their application is not a substitute to actual physicians. It remains in the importance of human check prior to making final medical decisions.

6.3.2 Improvements

This project can be enhanced in some few ways. First, we will be able to work with a larger and more diverse set of data of other hospitals or even countries. This will assist the model to learn more and provide more correct answers to varying kinds of people. Second, we are able to use additional more advanced steps such as cross-validation, feature selection, hyperparameter tuning to enhance model performance.

After this, we can attempt to use deep learning models powered by improved hardware such as GPU with a view to train CNN models in more effective ways. This should aid in improving precision as well as cutting down cost of training. Similarly, the data can also be cleaned even more by using additional data preprocessing techniques e.g., outlier detection or advanced scaling.

Finally, we may create user-friendly interface or a basic application in such a way that the model can be easily applied by the doctor or a nurse in real life. This will prevent the implementation of machine learning in real clinics and enhance early detection of heart diseases.

7. References

1. Hasani, W.S.R., Musa, K.I., Cheng, K.Y. *et al.* Exploring the trend of age-standardized mortality rates from cardiovascular disease in Malaysia: a joinpoint analysis (2010–2021). *BMC Public Health* **24**, 2519 (2024). https://doi.org/10.1186/s12889-024-19103-7
2. D'Agostino RB Sr, Vasan RS, Pencina MJ, Wolf PA, Cobain M, Massaro JM, Kannel WB. General cardiovascular risk profile for use in primary care: the Framingham Heart Study. Circulation. 2008 Feb 12;117(6):743-53. doi: 10.1161/CIRCULATIONAHA.107.699579. Epub 2008 Jan 22. PMID: 18212285.
3. Heart Matters: The Rising Burden of Cardiovascular Disease in Malaysia and Potential Touchpoints for Interventions. (2023). Malaysian Heart Association & IQVIA Asia Pacific. Retrieved July 13, 2025, from https://www.malaysianheart.org/management.file/doc/202312231133720341.pdf
4. Shinde, P., Sanghavi, M. & Tran, T.A. A Survey on Machine Learning Techniques for Heart Disease Prediction. *SN COMPUT. SCI.* **6**, 334 (2025). https://doi.org/10.1007/s42979-025-03860-2
5. Assegie, T. A. (2021). *Heart disease prediction model with k-nearest neighbor algorithm*. *International Journal of Informatics and Communication Technology (IJ-ICT), 10*(3), 225–230. https://doi.org/10.11591/ijict.v10i3.pp225-230
6. Huang, C., Li, X., & Deng, L. (2022). A deep convolutional neural network for the early detection of heart disease. *Computational Intelligence and Neuroscience*, 2022, Article 9687844. https://doi.org/10.1155/2022/9687844
7. Depar, S. A., Zaidi, S. M. J., Ali, M., Hashmi, M. U., Manan, A., & Ahmad, A. (2024). *A comparative study of machine learning models for heart disease prediction using grid search and random search for hyperparameter tuning*. *Journal of Computing & Biomedical Informatics, 8*(1). ZANA: Article 697 retrieved from https://jcbi.org/index.php/Main/article/view/697

8. Assegie, T. A., Sushma, S. J., & Mamanazarovna, S. S. (2023). Explainable heart disease diagnosis with supervised learning methods. *ADCAIJ: Advances in Distributed Computing and Artificial Intelligence Journal*, *12*(1), e31228. https://doi.org/10.14201/adcaij.31228 researchgate.net+4)
9. Chandrasekhar, N., & Peddakrishna, S. (2023). Enhancing heart disease prediction accuracy through machine learning techniques and optimization. *Processes, 11*(4), 1210. https://doi.org/10.3390/pr11041210
10. Jindal, R., Nayyar, A., & Rana, A. (2022). Machine learning-based heart disease prediction using logistic regression and improved K-means clustering. *Journal of King Saud University – Computer and Information Sciences*. https://doi.org/10.1016/j.jksuci.2022.04.002
11. Patel, T. S., Patel, D. P., Sanyal, M., & Shrivastav, P. S. (2023). *Prediction of heart disease and survivability using Support Vector Machine and Naïve Bayes algorithm.* bioRxiv. Retrieved from https://doi.org/10.1101/2023.06.09.543776